

Event Seating

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find
Domain: Integer Linear Programming

Universe

- Tables: $\{0, 1, 2\}$
- Model Stats: $\{total_variables, total_constraints\}$

Variables

Symbol	Meaning	Type / Range
x	decision variable	$\mathbb{R}$

Structure

Seat 12 wedding guests at 3 tables (4 guests each). Some pairs must sit together (couples). Some pairs must not sit together (feuding relatives). Maximize the total "affinity" score across all tables, where affinity measures how well pairs of guests get along.

Real-World Mapping

Real-World Concept	Mathematical Object
total score/cost	objective function value

Constraints

- | | | |
|-----|-------------------------------------|--|
| (1) | Constrained partitioning | dividing a set into groups subject to constraints |
| (2) | Integer linear programming | binary decision variables with linear constraints |
| (3) | Pairwise affinity modeling | encoding social preferences as an objective function |
| (4) | Hard constraints vs soft objectives | mandatory seating rules vs preference optimization |
| (5) | Linearization of quadratic terms | modeling "both at same table" with auxiliary variables |

Objective

$$\max \sum_{i,j} c_{ij} x_{ij}$$

Approach

Suggested approach: ILP (PuLP/CBC, Branch & Bound)

Complexity class:

Available tools: ILP

2. Solution

Answer:	Solution computed
Objective Value:	88.0
Optimal:	Yes
Feasible:	Yes
Algorithm:	ILP (PuLP/CBC, Branch & Bound)
Time:	0.6495s
Certificate:	Branch-and-bound solved to optimality.

Solution Details

Objective value: $z^* = 88.0$

Method: Integer Linear Programming (ILP) via PuLP/CBC. Binary decision variables $x[g,t]$ indicate whether guest g sits at table t . The objective maximizes the sum of pairwise affinity scores for guests seated at the same table. Constraints enforce: each guest at exactly one table, table capacity of 4, must-sit-together pairs at the same table, must-not-sit-together pairs at different tables.

Verification

✓ Feasibility	All constraints satisfied
✓ Optimality	ILP (PuLP/CBC, Branch & Bound)

3. Interpretation

The Question

Seat 12 wedding guests at 3 tables (4 guests each).

The Answer

The optimal solution has objective value $z^* = 88.0$.

What This Means

- Optimal seating arrangement for 3 tables of 4 guests each
- All must-sit-together constraints satisfied (couples at same table)
- All must-not-sit-together constraints satisfied (feuding pairs separated)
- Maximum total affinity score

Integer Linear Programming (ILP) via PuLP/CBC. Binary decision variables $x[g,t]$ indicate whether guest g sits at table t . The objective maximizes the sum of pairwise affinity scores for guests seated at the same table. Constraints enforce: each guest at exactly one table, table capacity of 4, must-sit-together pairs at the same table, must-not-sit-together pairs at different tables.

Recommendations

1. The solution is provably optimal; no further improvement is possible within this model.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.

• Solution